

Chapter 152 — Simple Prostatectomy: Open, Laparoscopic and Robotic Approaches

BACKGROUND/HISTORY

- **Impact of medical therapy:** medical therapy for benign prostatic hyperplasia (BPH) has profoundly changed management and narrowed surgical indications; endoscopic and interventional advances have added many options beyond transurethral resection.
- **Simple prostatectomy (enucleation) remains a time-tested gold standard** for bladder outlet obstruction from BPH in **glands >80 g**.
- **Simple vs radical:** simple prostatectomy removes **only the hyperplastic adenoma** and is performed for **benign indications**; radical prostatectomy removes the whole gland for cancer.

Perineal era

- **17th century:** first removals of adenomatous tissue — largely **inadvertent**, during open perineal lithotomy.
- **Gouley (New York), 1873:** described **total perineal enucleation**.
- **Zuckerkandl (Vienna), 1889:** elective extraurethral prostatectomy.
- **Goodfellow (American), 1890:** **earliest definitive series** of total perineal enucleation.
- **Hugh Hampton Young:** perineal technique allowing enucleation **under vision**.

Suprapubic and retropubic era

- **Atkinson and McGill (Leeds), 1887:** among the first **intentional suprapubic enucleations**, within days of each other.
- **Belfield (American)** presented his technique in 1890; **Fuller** in 1901.
- **Freyer (London):** popularised the **suprapubic (transvesical)** operation — initial series 1900, 1000th case by 1912, 1674th by 1920.
- **Millin, 1945:** developed the **retropubic approach without entering the bladder**.

Minimally invasive era

- **Laparoscopic simple prostatectomy (LSP) — Mariano, 2002 (first):** better recovery, pain and cosmesis, but blood loss still significant; **steep learning curve and poor ergonomics** limited diffusion.
- **Robotic-assisted simple prostatectomy (RASP) — Sotelo, 2008 (first, transperitoneal):** robotics eased the technical and ergonomic difficulty of pure laparoscopy; now widely performed.
- **Single-port (SP) laparoscopic transvesical prostatectomy** was described (2008) but technical difficulty prevented adoption; **SP robotic surgery** has since let the transvesical approach gain traction as **perhaps the least invasive, with the fastest recovery**, further reducing morbidity while keeping the advantages of robotics over pure laparoscopy.
- **Minimally invasive surgical (MIS) simple prostatectomy is evolving into the standard approach:** less pain, shorter stay, faster recovery, fewer complications — often **overnight stay or same-day discharge**.

Concept — *the history is one continuous reduction in access trauma for the same operation: the enucleation plane never changed; only the route to it did (perineal → suprapubic → retropubic → laparoscopic → robotic multiport → single-port transvesical).*

SURGICAL ANATOMY

Prostate Zonal Anatomy

- **McNeal zones:** peripheral, central and transition zone (TZ).
- **BPH arises in the TZ** → the TZ is the target of enucleation.
- As the TZ enlarges it **compresses the peripheral zone**, forming the **surgical capsule ("pseudocapsule")** — this is the **cleavage plane** of simple prostatectomy.

Pelvic and Vascular Anatomy

- **Main supply — prostatovesical artery**, most commonly from the **gluteopudendal trunk of the internal iliac artery**. Alternative origins: superior vesical artery; common trunk with the vesiculodeferential artery; internal pudendal or obturator artery.
- **Course:** runs medially on the levator ani to the bladder base → divides into **inferior vesical artery** and **prostatic artery**.
- **Prostatic artery** divides at the prostate base into:
 - **anterior branch** — anterolateral prostate;
 - **major posterolateral branch** — supplies **most of the gland**.
- **Proximal urethral branches** enter the capsule **posterolaterally just distal to the bladder neck** → supply the TZ (**the adenoma**). More distally, **capsular branches** supply the central and peripheral zones.
- **Venous drainage:** dorsal venous complex (DVC) → internal iliac veins.

Concept — *the adenoma's arterial supply enters posterolaterally at the bladder neck, which is exactly why the universal haemostatic step in every approach is a deep figure-of-eight suture at the 5 and 7 o'clock positions.*

INDICATIONS

- **Commonest indication:** BPH with gland >80–100 g and bothersome symptoms impairing quality of life, **refractory to medical therapy or prior endoscopic treatment**.
- **Medical sequelae of BPH/bladder outlet obstruction:** urinary retention; recurrent urinary tract infection; bladder stones; recurrent gross haematuria of prostatic origin; obstructive nephropathy.
- **Relative indication:** concomitant **bladder diverticula requiring surgery**.

PATIENT EVALUATION AND ANAESTHESIA

History and examination

- Full history, **International Prostate Symptom Score (IPSS)** and **postvoid residual (PVR)**.
- Medical conditions, medications, allergies and any **personal or family history of excessive bleeding**.
- **Prior abdominal/pelvic surgery** — adhesions or aberrant anatomy that dictate the route (intraperitoneal, extraperitoneal, transperineal or transvesical); **prior prostate surgery** complicates anatomy and dissection.
- **Digital rectal examination:** estimate size and **exclude nodules suspicious for occult cancer**.

Investigations

- **Laboratory:** prostate-specific antigen (PSA); metabolic panel and renal function; complete blood count (CBC) with platelets; coagulation profile; urine culture.
- **Positive culture** → **treat before surgery**; obtain sterile urine where possible.

- **Transrectal ultrasound (TRUS) strongly considered** — accurate size confirms simple prostatectomy is warranted rather than a less invasive procedure.
- **Cancer suspicion** (PSA, family history, examination, other risk factors) → **magnetic resonance imaging (MRI) and consider biopsy**.
- **Cystoscopy** as indicated → stones, diverticula, median lobe.
- **Urodynamics** → prolonged obstruction and retention with concern for an **atonic bladder**.

Preparation and anaesthesia

- Medical clearance; cardiology review if indicated. **Bowel preparation not routine**. Nil by mouth from midnight or per local protocol.
- **Preferred: general endotracheal anaesthesia**. **Spinal/epidural** acceptable for the **open approach only** — when general anaesthesia is contraindicated or by preference.

Concept — the work-up answers three questions: is the gland large enough to justify enucleation (TRUS), is there cancer that would make a simple operation the wrong one (examination, PSA, MRI), and is the bladder able to empty once unobstructed (urodynamics).

OPEN SIMPLE PROSTATECTOMY TECHNIQUE

Overview

Modern open simple prostatectomy (OSP) is **retropubic** (incision of the anterior pseudocapsule) or **suprapubic/transvesical** (extraperitoneal anterior vesicotomy, adenoma reached through a bladder neck mucosal incision).

RETROPUBIC VS SUPRAPUBIC OPEN SIMPLE PROSTATECTOMY

	Retropubic (Millin)	Suprapubic (transvesical)
Entry	Anterior prostatic pseudocapsule; bladder not opened	Anterior cystotomy; bladder neck mucosal incision
Advantages	Excellent exposure; direct, complete view of adenoma for total removal; precise distal urethral transection → sphincter protection ; best view of fossa for haemostasis; no bladder trauma	Simultaneously deals with large intravesical median lobe, clinically significant diverticula and large bladder calculi
Disadvantages	Venous bleeding from DVC disturbance; no bladder access for diverticula/calculi; poor access to a large intravesical median lobe	Bladder opened (cystotomy to close)
Choose when	Large gland, no intravesical pathology	Median lobe, diverticulum or stones

- **The deciding question is intravesical pathology:** anything needing the bladder open favours suprapubic.

Positioning and Initial Exposure

- **Cystoscopy** (if indicated): lithotomy first, or supine with a flexible cystoscope.
- **Supine, mild 15–20° Trendelenburg, no extension**. Suprapubic area shaved, prepped and draped; penis in the field, covered with a towel.
- **22-Fr (French) urethral catheter, 30-mL balloon**, placed sterilely. Right-handed surgeon stands on the patient's left.

ADENOMA ENUCLEATION

Retropubic — Millin Technique (Transverse Capsular)

Exposure

- Midline extraperitoneal incision (alternative: Pfannenstiel) → rectus retracted laterally → **transversalis fascia incised sharply → space of Retzius**.
- Posterior rectus fascia incised from the semicircular line to the umbilicus → **Balfour retractor**; peritoneum and bladder retracted cephalad with moist packs under a malleable blade.
- Periprostatic fat swept superolaterally without tearing veins; **superficial branch of the DVC cauterised and divided; endopelvic fascia incised bilaterally**, then puboprostatic ligaments transected bilaterally (Fig. 152.1).

Capsulotomy

- Bladder neck identified by palpation and catheter manipulation. Two stay sutures: one just distal to the bladder neck, one further distal.
- **Transverse capsulotomy ~2 cm distal to the bladder neck**, between the stay sutures, with a **No. 15 blade on a long handle**.
- **Why transverse, not longitudinal**: a longitudinal incision may **extend distally during retraction and endanger the external urinary sphincter**.
- Assistant compresses the proximal capsule (hand or sponge stick) to tamponade vessels and allow fulguration of the cut edge. Deepen to the adenoma by blade or cutting current; **the larger the gland, the further lateral the incision**.

Enucleation

- **Metzenbaum scissors find the plane between adenoma and pseudocapsule — entering the correct plane is essential** → index finger enucleates **laterally and posteriorly, easiest areas first**, adherent areas last.
- **Commissure-splitting method**: anterior commissure split base → apex (or capsulotomy carried through the adenoma to the urethra, catheter removed, commissure split caudally and cephalad from there) → **verumontanum seen directly** (with suction) or felt → left lobe divided at the apex **proximal to the external sphincter** → incision continued proximally in the posterolateral groove, **leaving a posterior urethral mucosal strip in continuity with the bladder neck** → left lobe out → right lobe likewise. The urethral mucosa is fractured with the index finger at the level of the verumontanum, with extreme care for the sphincter (Fig. 152.3).
- **Alternative (no commissure split)**: finger enucleation laterally and posteriorly → gland retracted cephalad → **prostatic-membranous urethral junction divided sharply under direct vision** → lateral lobes completed. No posterior strip is preserved, so the **posterior bladder neck lip may need excision** and is sutured down to the posterior capsule as distally as possible (2-0 chromic).
- **Median lobe**: bladder mucosa incised at the vesical neck under vision → median lobe enucleated (Fig. 152.4).

Haemostasis, catheter and closure

- Brisk bleeding → **pack the fossa for 5 minutes**.
- **Figure-of-eight 2-0 chromic or polyglactin sutures at 5 and 7 o'clock** through the proximal capsule; may include the bladder neck, **avoiding the ureteral orifices** → haemostasis **plus advancement of bladder neck mucosa into the fossa**. Verumontanum and posterior urethral strip remain intact (Fig. 152.5).

- Ureteral catheters/stents if orifices unusually close (rarely needed). Remaining visible vessels suture-ligated.
- **22-Fr three-way catheter, 30-mL balloon**, steered into the bladder with a long clamp before closure — or a **22-Fr two-way catheter plus a 22-Fr Malecot suprapubic tube**.
- **Closure: two running 2-0 absorbable sutures**, one from each lateral apex, meeting in the midline; tied first to themselves, then to each other → **watertight** (Fig. 152.6).
- **Retropubic drain**; irrigate the catheter to clear clots and test for leakage; **extra interrupted sutures** if leaking.

Retropubic — Vesicocapsular Technique

- **Described by Ogier Ward (1948), later Bourque (1954) and Watts (1956).**
- **Trade-off:** considered by some **easier than the transcapsular approach**, but **more urine leak** and **risk of caudal extension into the sphincter**.

Sequence

- Retzius developed → prostate and bladder depressed → **2-0 midline stay suture as distal as possible in the capsule** → two 2-0 stays either side of the bladder neck → **vertical incision through the bladder wall just above the prostate** → right index finger in the prostatic urethra, commissure broken → finger enucleation laterally, posteriorly, then bladder neck (as suprapubic) → capsule divided with curved Mayo scissors down to the distal stay → urethra divided at the apex under vision.
- **Haemostasis:** figure-of-eight 2-0 sutures at the **posterolateral positions (5 and 7 o'clock)**, plus sutures or cautery as needed. Protuberant, obstructive bladder neck edge → excised.
- Bladder neck mucosa advanced and tacked to the capsule over exposed detrusor (3-0 absorbable) → 22-Fr three-way Foley under vision → **3-0 submucosal purse-string from 4 to 8 o'clock** closes the vesicotomy → capsule closed with 2-0 absorbable, distal stay → proximal → detrusor and adventitia closed → balloon inflated, irrigated to check for leak.

Suprapubic Technique

Exposure and cystotomy

- Midline extraperitoneal incision, **pubic symphysis to umbilicus** → linea alba divided → Balfour → peritoneal reflection retracted cephalad → Retzius developed → bladder exposed.
- **Transverse cystotomy 1–2 cm proximal to the bladder neck** (electrocautery; Fig. 152.7); stay sutures above and below; **avoid capsular tearing** during stays, incision and stretching.
- **Vertical cystotomy** is an alternative but risks **distal extension** during finger enucleation or retraction.
- Bladder retracted cephalad with a padded malleable blade → **ureteral orifices identified** → **circumferential full-thickness bladder neck incision** around the adenoma, **distal to the trigone**, avoiding the orifices (Fig. 152.8).

Enucleation — two routes

SUPRAPUBIC ENUCLEATION: ANTERIOR VS POSTERIOR APPROACH

Anterior-focused	Posterior-based
Finger in prostatic urethra → anterior pressure "fractures" the anterior commissure	Plane developed at the posterior midline by spreading Metzenbaum scissors (Fig. 152.9)
Finger swept laterally round left and right lobes → across the posterior midline	Index finger (surgeon's back to the feet) carries the plane circumferentially and distally to the apex (Fig. 152.10)

Anterior-focused	Posterior-based
Both end the same way: apex retracted cephalad → prostatic urethra pinched through between thumb and forefinger — no excessive traction (sphincter) → adenoma removed whole or by lobes with sponge forceps. Extremely large glands: remove left, right and median lobes separately (Fig. 152.10).	

Haemostasis and closure

- **Fossa packed, Deaver pressure 5 minutes** → fossa contracts; inspect and remove residual tissue.
- **Deep figure-of-eight 2-0 chromic at 5 and 7 o'clock within the capsule** → controls prostatic arteries; small bites of bladder mucosa **advance it into the fossa over exposed detrusor to prevent contracture**. Further bleeding: cautery or suture ligatures.
- **Catheter: 22-Fr three-way, 30-mL balloon** — or 22-Fr two-way plus **20–24-Fr Malecot suprapubic tube**.
- **Cystotomy closed in two layers: 3-0 Vicryl mucosa, 2-0 Vicryl muscle and adventitia** (Fig. 152.11).
- **Balloon filled with 50 mL** → prevents retraction into the fossa, especially if traction becomes necessary; irrigate to clear clot and check for leak.
- Closed-suction or Penrose drain via a **separate stab incision** lateral to prostate and bladder; pelvis irrigated; **rectus fascia closed with No. 1 polydioxanone (PDS), running**; skin closed.

Concept — every open variant is the same four moves — enter, find the pseudocapsular plane, divide the urethra at the apex without touching the sphincter, then 5/7 o'clock haemostasis with bladder neck mucosal advancement — and the variants differ only in where the entry incision sits and therefore what it risks (DVC bleeding retropubically, leak and sphincter extension vesicocapsularly, nothing intravesical missed suprapubically).

MANAGEMENT OF HAEMORRHAGE

Intraoperative Haemostatic Manoeuvres

INTRAOPERATIVE HAEMOSTATIC TECHNIQUES

Technique	When	Method
Walsh and Oesterling (1990) — prophylactic vascular control	Before retropubic (Millin) enucleation	Superficial dorsal vein cauterised → endopelvic fascia incised → partial puboprostatic ligament transection → DVC ligated (3-0 Monocryl, 5/8 tapered needle) in the avascular plane between urethra and DVC at the apex → lateral pedicles at the prostatovesical junction (2-0 chromic, CTX needle, figure-of-eight) deep where the seminal vesicles meet the prostate, avoiding entrapment of the posterolateral neurovascular bundles (Fig. 152.12)
Standard 5 and 7 o'clock sutures	Every case	Usually sufficient with or without prophylactic control
11 and 1 o'clock sutures	Occasionally	Anterior arterial branches
Malament (1965) purse-string partition	Pronounced, continued bleeding	No. 1 or 2 double-armed nylon round the vesical neck (mucosa and muscle), from 6 o'clock outside, arms clockwise and anticlockwise to 12 o'clock, not crossed (else irremovable) → full-thickness through bladder → 24- or 26-Fr Foley, 30-mL balloon → purse-string drawn tight round catheter → ends through the abdominal wall above the symphysis, tied under tension over a button/pledget → bladder neck closed and fossa compressed and partitioned from bladder; Malecot suprapubic tube and retropubic drain; suture cut at skin and withdrawn on day 2–3

Technique	When	Method
O'Connor (1982) capsular plication	Bleeding without an obvious source	Two transverse plicating sutures (No. 1 chromic, 5/8 needle) in the posterior capsule/fossa — one near the bladder neck, one distal; tying approximates the capsule → compression

Postoperative Haemostatic Manoeuvres

- **Escalation:** catheter traction with 50 mL in the balloon (prevents migration into the fossa) ± **continuous bladder irrigation (CBI)** → **exclude coagulopathy** (laboratory testing) → failure of conservative measures → **return to the operating room (OR)**.
- **In the OR:** endoscopic first — resectoscope for **clot evacuation and cauterisation**; **severe haemorrhage** → **reopen wound and prostatic fossa**.

Concept — *haemostasis is graded by escalation of the same target: suture the arterial inflow (5/7, then 11/1 o'clock), then compress the fossa mechanically (plication, purse-string partition, balloon traction), and only then return to theatre.*

POSTOPERATIVE CARE

- **Recovery unit:** vital signs, drain and catheter outputs; CBC if bleeding concern.
- Catheter traction and/or CBI as needed; **sequential compression stockings**; **non-opioid analgesia**, opioids for breakthrough only.
- **Day 1:** diet advanced; **incentive spirometry and early ambulation**.
- **CBI weaned as haematuria allows**; balloon partially deflated to 30 mL to reduce irritation and spasm; hand irrigation if clots.
- **Day 2, urine clear:** remove Foley, **clamp suprapubic tube (SPT) for trial of void**, discharge. **No SPT** → discharged with the Foley.
- **Pelvic drain** removed before discharge if output low; **high output** → **drain creatinine** to exclude urinary leak.
- Ambulate at home to reduce venous thromboembolism.
- **1 week:** Foley removed; or SPT removed if voiding comfortably with minimal PVR.
- **No heavy lifting ~4 weeks.** Follow-up at **6 weeks and 3 months:** symptoms and PVR.

MULTIPORT ROBOTIC-ASSISTED SIMPLE PROSTATECTOMY TECHNIQUE

Overview

- **Multiport robotic-assisted simple prostatectomy (MP-RASP)** is most often transperitoneal; extraperitoneal is an alternative. Retropubic or suprapubic techniques each **replicate the open steps**.

Patient Positioning and Operating Room Setup

- Supine on a **Pink Pad** (or similar) so Trendelenburg causes **no slippage**; endotracheal intubation, **orogastric tube**; arms tucked, pressure points padded.
- **Compression stockings plus pharmacological thromboprophylaxis.**
- **Trendelenburg tested before prepping** — confirms cardiopulmonary tolerance and no movement.
- Penis prepped in field; **18-Fr Foley, 10 mL in balloon.**

Access and Trocar Placement

- **Transperitoneal:** Veress (or Hasson) → insufflate to **15 mm Hg** → **8-mm umbilical camera port** (da Vinci Xi) → inspect for access injury and adhesions → two 8-mm arm ports **8–10 cm lateral** to camera → fourth-arm 8-mm port **8 cm right of the right port** → **5-mm and 12-mm assistant ports** (layout may be mirrored; Fig. 152.13) → operate at **8–12 mm Hg**.
- **Extraperitoneal:** 2-cm infraumbilical incision → vertical incision in anterior rectus sheath → split recti → **balloon dilator towards Retzius** → ports as above, shifted **~2 cm caudally**, below the camera.

Multiport Transperitoneal Suprapubic Approach

- Pneumoperitoneum **8–12 mm Hg**; 0° or 30° scope. **Instruments:** bipolar Maryland (left), monopolar curved scissors (right), ProGrasp (fourth arm).
- Saline via Foley defines the bladder → **vertical midline cystotomy** (horizontal acceptable), bladder emptied before mucosal entry; thick hypertrophic wall divided with monopolar and bipolar cautery.
- **Retraction:** 3-0 Vicryl or barbed stay sutures — anterior and posterior on each side (typically four; Fig. 152.14), through adventitia and detrusor then laterally through peritoneum, held with Hem-o-Lok clips; **avoid inferior epigastric vessels (anterior) and external iliac vessels (lateral)**. Alternative: straight Keith needle through abdominal wall and bladder, clamped at the skin.
- Cystotomy extended for a full view of the bladder neck; **ureteral orifices identified**.
- **Advantages of direct dome entry:** fast entry to the bladder; **any capsular perforation stays in the unviolated extraperitoneal space**.
- **Alternative — "drop" the bladder** as in anterior radical prostatectomy (no endopelvic fascia incision) → transverse full-thickness incision **1–2 cm proximal to the bladder neck**, extended laterally.

Enucleation sequence

- Start **posteriorly under the intravesical lobe:** bladder neck incision **~3 → ~9 o'clock**, full thickness to the adenoma–pseudocapsule plane, **well away from ureteral orifices** (catheterise orifices if obstruction concern; rarely needed; Fig. 152.15).
- Median lobe lifted by fourth-arm tenaculum or retraction suture; **large intravesical median lobe may be excised early** (Fig. 152.16).
- Posterior plane developed bluntly to the apex (30° up scope helps) → bladder neck incision completed circumferentially.
- Right lobe dissected **anticlockwise** (adenoma retracted medially, pseudocapsule swept laterally; bipolar only for pulsatile vessels; scissors if adherent) → over the apex (Fig. 152.17) → left lobe **clockwise**.
- Before apical transection: **anterior commissure divided and verumontanum visualised** (bedside suction key) → **urethra divided just proximal to the veru**, never distal (sphincter; Fig. 152.18) → left then right lobe enucleated; fourth arm toggles the gland throughout → specimen bag.

Haemostasis, reconstruction and closure

- **Figure-of-eight 3-0 barbed at 5 and 7 o'clock**, deep in posterior capsule; bleeders bipolar-cauterised or oversewn. **Avoid excess cautery, especially posterolaterally — thermal injury to neurovascular bundles → erectile dysfunction**.
- **Vesicourethral mucosal advancement:** double-armed 3-0 barbed, **outside-in at 6 o'clock**, one arm clockwise, one anticlockwise, tied anteriorly (Fig. 152.19) → **urothelial continuity, faster healing, haemostasis by excluding the oozing fossa** → patients typically leave theatre **without CBI**.
- **22- or 24-Fr three-way Foley, 30-mL balloon;** bladder closed in **two layers: 3-0 barbed mucosa, 2-0 barbed detrusor/adventitia**; optional third layer of peritoneum and perivesical fat (Fig. 152.20). **No routine SPT or drain**.

- Specimen out through the umbilical incision; **home day 1 with Foley, removed at 5–7 days.**

Multiport Transperitoneal Retropubic (Millin) Approach

- Bladder dropped into Retzius → stay sutures at both apices → **transverse capsulotomy** deepened with cautery to the adenoma → blunt and cautery dissection laterally and posteriorly → urethra incised distally → adenoma out **en lobe or en bloc** → haemostasis as above → **bladder neck mucosa to urethral mucosa** (double-armed 3-0 barbed) → capsulotomy closed with 2-0 barbed or polyglactin.

Multiport Extraperitoneal Approach

- Preperitoneal space made by finger and balloon → **longitudinal cystotomy from anterior bladder wall to bladder neck** → adenoma removed as transperitoneally.
- **Advantages:** direct access to the vesicoprostatic junction; **avoids the peritoneum — for the hostile abdomen** after complex prior surgery.

Concept — *the robotic operation adds one step the open operation lacks — the vesicourethral mucosal advancement that re-covers the fossa — and that single step is what removes the need for irrigation, drain and suprapubic tube.*

SINGLE-PORT ROBOTIC SIMPLE PROSTATECTOMY TECHNIQUE

Overview

- **Transperitoneal single-port robotic-assisted simple prostatectomy (SP-RASP)** — easiest transition from multiport: straightforward transperitoneal access; simple bladder incision and closure; large working space; option of **"plus-one" accessory ports** early in the learning curve.
- **Transvesical SP-RASP = single-port transvesical enucleation of the prostate (STEP)** — **least invasive suprapubic prostatectomy**: one **2.7-cm** incision, **no peritoneal entry, pneumovesicum** (not insufflation of peritoneum or extraperitoneal space) → least pain, rapid recovery, best cosmesis, short catheter time, **no drains, little or no CBI, often same-day discharge** — competitive with **transurethral laser enucleation for large glands.**

Single-Port Transperitoneal Suprapubic Approach

- Supine, penis in field, **18-Fr Foley** → **2.7-cm vertical transumbilical incision** → peritoneum opened sharply under vision → da Vinci SP Access Port Kit → **8–12 mm Hg** → Trendelenburg → dock.
- **Instrument positions:** camera **12 o'clock**; Maryland bipolar left; monopolar scissors right; fenestrated bipolar or Cadere **6 o'clock** (retraction and handling the suction). **Remotely Operated Suction Irrigation (ROSI) is essential for vision.**
- Steps as MP: bladder entry → bladder neck incision → enucleation → running vesicourethral apposition → bladder closure → specimen out → fascial closure. **No routine SPT/drain; home day 1, Foley out day 5–7.**

Single-Port Transvesical Approach (STEP)

Access

- 18-Fr Foley → bladder filled with **~200 mL saline** → **transverse 2.7-cm incision two fingerbreadths above the pubic symphysis** (Fig. 152.21) → fascia opened vertically, recti retracted → perivesical fat spread (Kelly) → bladder confirmed by **Veress/18-gauge needle with urine/saline return.**
- Four 2-0 polyglactin stays (proximal, distal, left, right) → **3-cm vertical full-thickness cystotomy** → bladder emptied → SP Access Port → dock (Fig. 152.22) → **pneumovesicum 8 mm Hg.**

Enucleation

- Instruments as SP transperitoneal; orifices identified; **semicircular bladder neck incision ~3 → ~9 o'clock**, well away from orifices, full thickness to the plane. **Large intravesical lobes may be excised first.**
- **Posterior plane is the most distinct — exploit it early and as distally as possible**, bluntly to the apex (Fig. 152.23). To see it: relocate the SP system posteriorly and angle the camera anteriorly, or move the camera to 6 o'clock.
- Then clockwise and anticlockwise along the adenoma–pseudocapsule interface, **incising the bladder neck as you advance**; adenoma retracted medially, pseudocapsule swept laterally; some sharp dissection and cautery (Fig. 152.24).
- **Very large glands:** divide the anterior commissure → lobes separately; otherwise **en bloc**. **Veru seen directly** → urethra and apex transected without sphincter injury. Specimens parked in the balloon port.

Haemostasis and closure

- **5 and 7 o'clock figure-of-eight 3-0 barbed**; bleeders cauterised or oversewn.
- **Double-armed 3-0 barbed, outside-in on the bladder neck side at 6 o'clock**, arms clockwise/anticlockwise → bladder neck descends → **mucosa-to-mucosa anastomosis**, tied anteriorly (Fig. 152.25).
- **22-Fr three-way Foley, 30-mL balloon filled with 20 mL**; bladder checked for **residual fragments (can obstruct the catheter)**; port and specimens out; **two-layer closure 3-0 and 2-0 absorbable under direct vision** (Fig. 152.26); **no drain or SPT**; irrigate to confirm watertightness.
- **Rarely needs CBI; home same day or morning of day 1 with Foley; Foley out day 2.**

LAPAROSCOPIC SIMPLE PROSTATECTOMY TECHNIQUE

- Positioning and anaesthesia **as for MP-RASP**; Veress or Hasson access.
- **Ports:** 10-mm umbilical camera; two 10-mm working ports **8–10 cm lateral**; 5-mm port inferolateral to the 10-mm on the surgeon's side; **12-mm AirSeal** contralaterally.
- Steps as MP-RASP: either **drop the bladder and make a horizontal incision 2 cm proximal to the bladder neck**, or **enter the bladder directly without entering Retzius** → mucosal incision, enucleation, haemostasis, closure.

COMPARISON ACROSS APPROACHES — ACCESS, CATHETER AND DISCHARGE

Approach	Access	Catheter	Drain / SPT / CBI	Discharge; catheter out
Open retropubic / suprapubic	Lower midline (Pfannenstiel alternative retropubically)	22-Fr three-way, 30 mL (suprapubic balloon filled to 50 mL); or two-way + Malecot	Drain routine; SPT optional; CBI often	Foley out / SPT trial day 2 if clear; otherwise home with Foley, out at 1 week
MP-RASP	6 ports; transperitoneal or extraperitoneal	18-Fr at start; 22–24-Fr three-way, 30 mL at end	Not routine ; CBI rarely	Day 1; Foley out 5–7 days
SP transperitoneal	2.7-cm umbilical	18-Fr at start	Not routine	Day 1; Foley out 5–7 days
STEP (SP transvesical)	2.7-cm suprapubic; pneumovesicum 8 mm Hg	22-Fr three-way, 20 mL in 30-mL balloon	None ; CBI rare	Same day or day 1; Foley out day 2

Approach	Access	Catheter	Drain / SPT / CBI	Discharge; catheter out
LSP	10-mm camera, two 10-mm, 5-mm, 12-mm AirSeal	As MP-RASP		

- **Pattern:** the less invasive the access, the shorter the catheter time — STEP is the only approach with catheter out on day 2 and same-day discharge.

OUTCOMES AND COMPLICATIONS

Efficacy and haemorrhage

- **OSP is efficacious and durable:** high satisfaction, better quality of life, objective improvement in maximum urinary flow rate (Qmax) and PVR.
- **Historically, haemorrhage was the major concern**, with high morbidity and transfusion rates; with modern technique **open transfusion <5%**. Laparoscopic and robotic approaches have reduced bleeding further — **transfusion essentially eliminated**, and **CBI for routine fossa oozing no longer routinely required**.

COMPARATIVE OUTCOMES BY APPROACH (BOTTOM LINES)

Comparison	Better / different	Equivalent
LSP vs OSP	LSP: less blood loss, shorter irrigation, catheter time and stay, less analgesia ; lower UTI	Complication rates, functional outcome
Robotic vs LSP	Robotic: better ergonomics, easier learning curve — has largely supplanted LSP	Efficacy not superior ; complications, transfusion, incontinence, IPSS and sexual function (Sexual Health Inventory for Men) at 1 year all similar
Robotic (RSP/RASP) vs OSP	Robotic: less blood loss and transfusion, shorter stay and catheter time, fewer recatheterisations, UTIs, readmissions and overall complications (including respiratory, ileus, sepsis, genitourinary); but longer operating time	IPSS, Qmax, PVR, urodynamics, quality of life
SP (STEP) vs MP	SP: shorter stay and catheter time, far less CBI, lower opioid use (most need none after discharge), fewer readmissions; same-day discharge for most SP patients, none after MP	Operating time, blood loss, major complications, reoperation

- **Initial STEP experience:** minimal blood loss, **stay under one day**, no intraoperative complications, even early in the learning curve.
- **Consistent pattern:** minimally invasive routes improve **perioperative morbidity**; **functional outcome is the same for all** — because the enucleation is the same.

Urinary extravasation

- **Uncommon.** Sources: incomplete closure of the **capsulotomy (retropubic)** or **cystotomy (suprapubic)**; **unrecognised inadvertent capsulotomy during suprapubic prostatectomy**.
- **Recognition:** high drain output with **drain creatinine consistent with urine**. Rare presentations: **urinary ascites or urinoma** ± infection.
- **Management:** **leave drain and Foley until it stops** — spontaneous resolution with time.

Ureteral orifice injury

- Direct injury or **oedema-related obstruction** from the bladder neck incision, haemostatic sutures, or bladder neck-urethra reapproximation.

Infection

- **Relatively rare:** cystitis, epididymitis, infected collection. LSP had lower UTI than OSP; **robotic vs open meta-analysis showed no UTI difference.**

Bladder neck contracture (BNC)

- **OSP: 2–6%. MIS series: <1%** in most; robotic and open were comparable in one comparison.
- **Management:** dilation with urethral sounds or **direct-vision incision with a Collings knife to a 22-Fr opening.**

Non-urological complications

- **Deep vein thrombosis, pulmonary embolus, myocardial infarction, stroke** — each <1%; **mortality should approach zero.**

RECOVERY AND SIDE EFFECTS

- **Storage symptoms** (frequency, urgency, urge incontinence) may last **weeks to months**; severe → **anticholinergic or β_3 -adrenergic receptor agonist.**
- **Stress and total incontinence: extremely rare** — precise enucleation leaves the **external sphincter** untouched; robotic series report urinary incontinence in <6%.
 - **Mild stress incontinence** → **transurethral collagen injection**; **severe** → **artificial urinary sphincter.**
- **Erectile dysfunction: 3–5%**, commoner in older men; in robotic series, **erectile function usually unchanged or improved.**
- **Retrograde ejaculation: 80–90%** — expected.

Concept — *the adenoma lies inside the pseudocapsule and proximal to the verumontanum, so removing it destroys the bladder neck mechanism (retrograde ejaculation is near-universal) but spares the external sphincter and the neurovascular bundles outside the capsule (incontinence and erectile dysfunction are rare).*

KEY POINTS

- Consider simple prostatectomy when obstructing tissue is **>80 g** or when **sizable bladder diverticula or calculi** exist.
- **Investigate for significant prostate cancer before surgery.**
- **Retrograde ejaculation is common and expected**; urinary incontinence and erectile dysfunction are **very low risk.**
- **Complications:** UTI, bladder neck contracture, urethral stricture, deep vein thrombosis, pulmonary embolus, transfusion.
- **Advantages over transurethral resection of the prostate (TURP):** **lower retreatment rate**, more complete removal under direct vision, **no risk of transurethral resection (TUR) syndrome.**
- **Disadvantages of OSP over TURP:** lower midline incision, longer hospitalisation, more perioperative haemorrhage.
- **Robotic vs OSP:** smaller incisions, shorter catheterisation, much less haemorrhage and transfusion, shorter stay, often same-day discharge.
- **Modern open approaches:** **retropubic** (transcapsular Millin or vesicocapsular) and **suprapubic (transvesical).**

- **Suprapubic preferred over retropubic for a large median lobe, clinically significant diverticulum and/or large bladder calculi.**

SIMPLE PROSTATECTOMY VS TURP

Simple prostatectomy advantages	Open simple prostatectomy disadvantages
Lower retreatment; complete removal under vision; no TUR syndrome; handles diverticula/stones; suits men who cannot be placed in lithotomy	Lower midline incision; longer stay; more perioperative haemorrhage — the disadvantages that MIS approaches remove

SUMMARY

- Open, MP or SP robotic simple prostatectomy is an excellent option for markedly enlarged BPH glands with: **(1) symptomatic outlet obstruction; (2) medical sequelae of obstruction; (3) concomitant bladder conditions** (diverticula, large calculi); **(4) inability to adopt dorsal lithotomy for TURP.**
- **Proven, durable efficacy** — improved symptom score, peak flow and PVR — with minimal complication risk, including haemorrhage.
- Technology now allows routine surgery **without continuous irrigation, with short catheter duration, even with same-day discharge.**